# Deepak Mallampati


**APPLIED AI ENGINEER - AI PLATFORM / LLM OPS & ML OPS**

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Kent, OH - US-based, F-1 OPT (open to sponsorship)


## PROFESSIONAL SUMMARY


Applied AI / ML engineer with 4+ years building and operating production LLM and ML systems, with a platform focus on LLM Ops and ML Ops - model access, evaluation, monitoring, deployment, governance, and cost/latency control. Built a structured LLM evaluation and monitoring framework (RAGAS, BERTScore, latency/accuracy dashboards) that surfaces regressions before release, and engineered inference-cost systems (semantic caching, model routing, prompt compression) that held SLA while cutting per-query overhead. Delivered reusable AI platform building blocks - 25+ agents and 12 APIs on a shared internal platform serving 100k+ requests/day across 20+ teams - with 99.9% uptime and enterprise-grade safety/governance controls.


## CORE COMPETENCIES

LLM Ops • ML Ops • Model Observability & Monitoring • LLM Evaluation (RAGAS, BERTScore) • Model Routing & Semantic Caching • Deployment & Governance • Developer Platforms & Internal Tooling • RAG Architecture • Multi-Agent Systems

**MLOps & Platform:** MLflow, Weights & Biases, Model Monitoring & Observability, model routing, semantic caching, Docker, Kubernetes, Terraform, CI/CD (Jenkins, Azure DevOps), FastAPI microservices

**Generative AI & LLMs:** GPT-4o/GPT-5, Claude Sonnet, Gemini, LLaMA, LangChain/LangGraph, RAG, prompt engineering, multi-agent systems, fine-tuning (LoRA/QLoRA/PEFT), Content Safety, Hugging Face Transformers

**Machine Learning & NLP:** PyTorch, TensorFlow, Keras, NER, text classification, summarization, model optimization, A/B testing, drift detection

**Vector Search & Retrieval:** FAISS, Pinecone, Weaviate, semantic & hybrid search, cross-encoder re-ranking, chunking strategies

**Cloud & Data:** AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure (OpenAI, AI Services), GCP Vertex AI, PostgreSQL, MongoDB, Redis, Kafka, Spark, Databricks

**Security & Compliance:** HIPAA, SOC 2, PCI DSS, OAuth 2.0/JWT/RBAC, PII masking/redaction, prompt-injection mitigation, audit logging

**Languages & Tools:** Python, JavaScript/TypeScript, Java, SQL, FastAPI, React, Git, Jira, Confluence, Agile/Scrum

## PROFESSIONAL EXPERIENCE

**Progress Solutions** — *Jul 2025 - Present | USA*

*AI Engineer*

- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) paired with latency and accuracy dashboards to benchmark model iterations pre-deployment and surface regressions before release - a reusable platform primitive other teams build on.
- Optimized inference **cost and latency** through prompt compression, semantic caching, and **model routing**, sustaining SLA targets while reducing per-query overhead.
- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**.
- Engineered high-performance **RAG** pipelines over large healthcare corpora (dense retrieval + cross-encoder re-ranking) to lift answer relevance and materially reduce hallucination.
- Operationalized **privacy-preserving data workflows** (K-anonymity, L-diversity, differential privacy) for compliant analytics with zero PII exposure.

**Vanguard** *Jan 2024 - Jan 2025 | USA*

*AI Engineer Intern*

- Developed **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and **observability tooling** across **25+ AI agents** and workflows.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by 40%.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by 27% and informing platform model-selection decisions.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining response quality.
- Partnered with product, platform, and data-engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

**Kent State University** *Oct 2023 - Jan 2024 | USA*

*ML & Gen AI Research Assistant*

- Built a **privacy-preserving ML pipeline** on a clinical readmission dataset, cutting re-identification risk from 3.38% to 0.32% while retaining 99% of baseline predictive performance.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 with a curated instruction dataset for controllable long-form generation; authored an IEEE-format conference paper.

**Emerson** *Jun 2022 - Apr 2023 | India*

*ML Trainee*

- Trained supervised **regression and classification models** for equipment-failure prediction, anomaly detection, and capacity forecasting on operational sensor data.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving 89% F1 on custom NER tasks.

## PROJECTS

**career-ops**

An autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision - end-to-end orchestration with logging, retries, and observability baked in.

**MangaLens**

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across 29+ sites - a production GenAI product turning a multi-step manual chore into a single inline action.

**Privacy-Preserving Clinical ML Pipeline**

A framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive patient data without exposing identities, quantifying the privacy-utility trade-off of each technique.

## EDUCATION

**M.S. in Computer Science -** *Kent State University, OH* *2025 | GPA 3.70/4.0*